Report on Predicting Resale Prices of HDB Flats

1. Introduction

This analysis aims to predict HDB resale prices in Singapore using a dataset of flat characteristics. Buyers and sellers can benefit from this, aiding in better decision-making, by understanding factors and trends which affect resale values. Initially, Linear Regression will serve as a baseline, followed by Decision Tree, k-NN, and PCA models. Model performance will be evaluated using Root Mean Squared Error (RMSE), which measures the average magnitude of prediction errors by calculating the square root of the average squared differences between predicted and actual values, offering interpretability in the original price units. The Margin of Error, based on the mean price of $497,485.70, will further assess model accuracy. The dataset was reduced from 230 to 133 predictors to minimise redundancy.

2. Data Processing

Feature Transformation

The resale price variable was right-skewed (shown in Figure 1), with high-value outliers (Appendix A), which could negatively affect model performance. To address this, I applied a log transformation to the resale price, helping to stabilize variance and make the distribution more symmetric (Shown in Figure 2). This transformation aligned the data with the assumptions of many statistical models, reducing the impact of extreme values. The transformation is as shown below in the histograms.

Figure 1: Initial Resale Prices (Right Skewed)

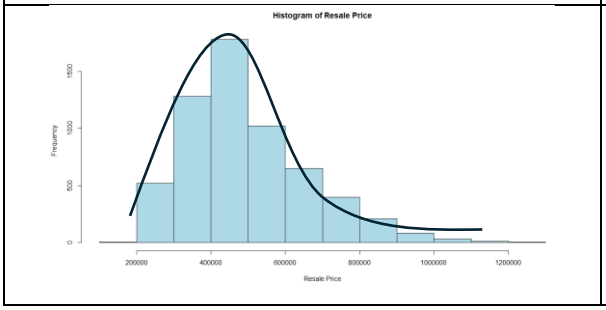
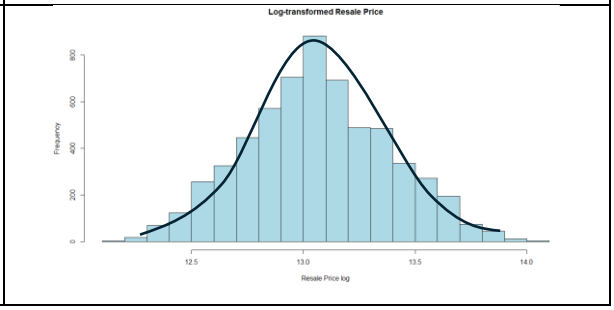


Figure 2: log-transformed Resale Prices



3. Model Testing

For all models, 10-fold cross validation (CV) was used to evaluate its performance. This method helps provide a more reliable estimate of the model's performance, staying consistent with all models. It also helps to improve effectiveness of the model on unseen data. A 4000/2000 train-test split was made.

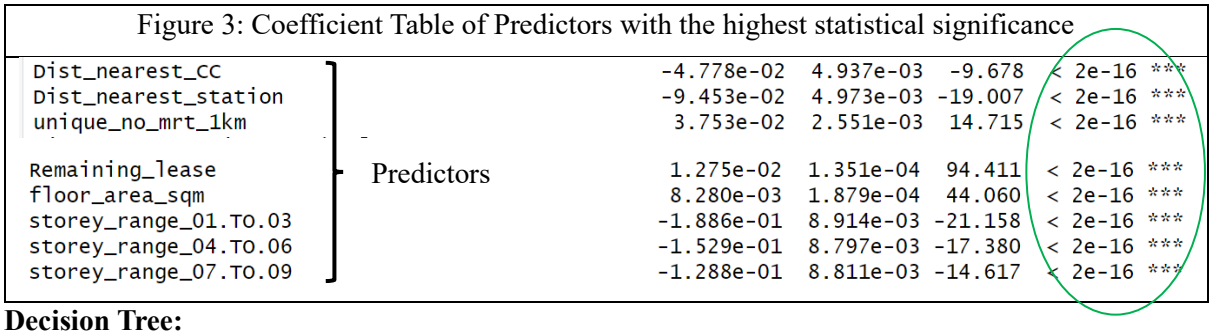
Linear Model (LM):

The linear model was fit using forward selection, resulting in a refined set of 91 predictors. The model was then trained using these predictors, and its performance was evaluated on the test set, calculating the RMSE on the original scale.

Adjusted R-Square (Before CV)	0.927
RMSE (Before CV)	53,956.25
Adjusted R-Square (After CV) – Best Model	0.9246
RMSE (After CV) – Best Model	51,450.73

I observed that while the RMSE improved, the Adjusted R-squared value decreased slightly, indicating a trade-off in model fit for better predictive performance. Key predictors shown in figure 3 had the highest contributions based on the significantly low p-value, indicating a very low probability of having no connection between the predictor and the response variable. These are typical considerations for buyers, supporting the reliability and practical relevance of my linear model.

Figure 3: Coefficient Table of Predictors with the highest statistical significance



Decision Tree:

A decision tree model was built using the "rpart" package to identify the best complexity parameter (cp) value. The best cp, which balances tree complexity and performance, was determined by plotting the cross-validation error graph (Appendix B) and choosing based on the minimal RMSE. The kernal chosen was “gaussian”, which prioritises closer points to the given point, adapting to noise in the data.

Best cp from Cross-Validation:	0.000112338
RMSE (Before CV)	55104.15
RMSE (After CV) – Best Model	55071.20
Unique Terminal Nodes Initial (Leaves):	226

To prioritise accuracy, I built a tree model with 226 leaves, sacrificing some interpretability for performance. This model was competitive with the Linear Model when comparing RMSE. Although it may perform slightly worse, the tree's resilience to collinearity and scaling showcased its robustness. This made it a computationally efficient and swift model to construct due to no pre-processing required.

K-Nearest Neighbours (k-NN):

The k-NN model predicts resale prices by finding similar properties and averaging their prices. Cross-validation was used to test various k values, the number of nearest neighbours for a given data point, to identify the best k based on the minimal RMSE. Best k value chosen is 3.

RMSE (Before CV)	74540.55
RMSE (After CV) – Best Model	75195.51

I tested the model with a range of k values, but the accuracy remained stagnant, prompting a reduction from 1:50 to 1:20 for computational efficiency. The k-NN model's RMSE was significantly higher than other models, making its prediction worse than other models tested.

Principle Component Analysis (PCA):

PCA simplifies complex data by emphasizing key patterns and removing redundant information, creating a smaller set of transformed variables. Features with zero variance were excluded prior to applying PCA. The scree plot (Appendix D) guided the selection of optimal components for analysis. These components were then used to develop a Principal Component Regression (PCR) model.

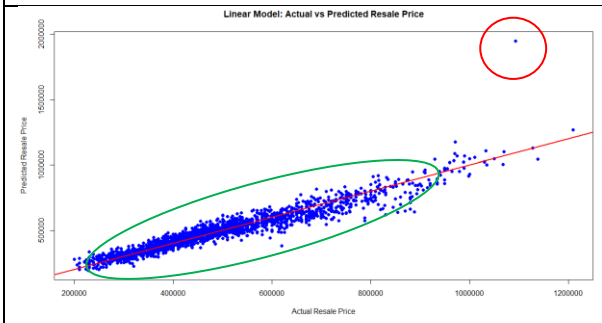
Number of Components Chosen:	69
RMSE	40092.43

To enhance prediction accuracy, I tuned the PCA model to select variables that explained 95% of the variance. This reduced the data's dimensionality from 133 to 69 predictors, effectively halving the count. The model achieved the lowest RMSE compared to others and an adjusted R-Square of 0.9487, showing its closeness in perfect fit of data and efficiency in reducing noise.

4. Model Evaluation and Comparison

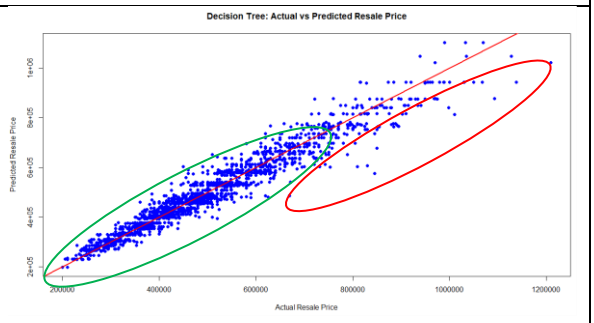
Actual vs Predicted Resale Prices - All Models **(Straight Line Graph (Red): Perfect Prediction)**

Figure 4: Linear Model



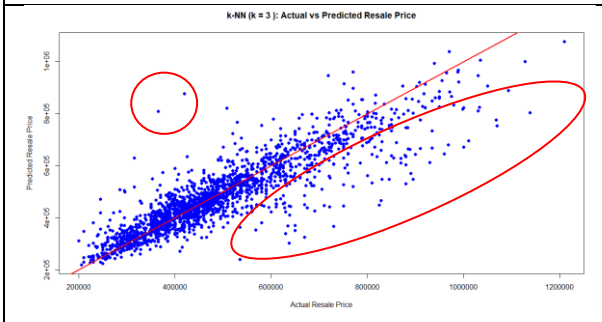
Margin of Error: $\frac{51450.73}{497485.68} \times 100 = 10.34\%$

Figure 5: Decision Tree



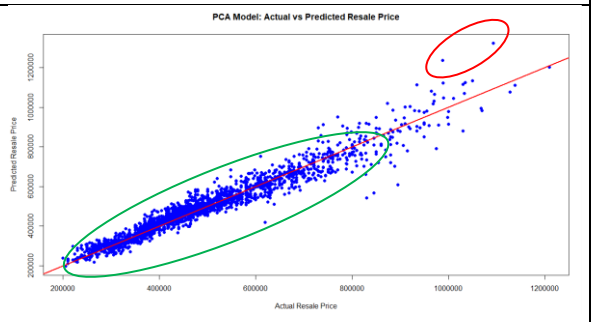
Margin of Error: $\frac{55071.20}{497485.68} \times 100 = 11.07\%$

Figure 6: Knn



Margin of Error: $\frac{75195.51}{497485.68} \times 100 = 15.12\%$

Figure 7: PCA



Margin of Error: $\frac{40092.43}{497485.68} \times 100 = 8.06\%$

In this analysis, we compare the performance of the four models, based on their margin of error and graphical evaluations. The Linear Model has a margin of error of 10.34% and shows closely knit points near the perfect prediction line at lower prices, with slight scatter and one extreme anomaly at higher prices. The Decision Tree, with an 11.07% margin of error, follows a similar pattern but underpredicts more at higher prices. The k-NN model, which has the highest margin of error at 15.12%, displays more scatter overall, especially beyond the mean price where underpredictions dominate. In contrast, the PCA model has the lowest margin of error at 8.06%, with most points tightly aligned to the perfect prediction line and only a few outliers that are not significantly distant, making it the most accurate model.

5. Conclusion

In this analysis, various models were tested to predict the resale prices of HDB flats, focusing on accuracy. The Linear Model and Decision Tree performed well, with the Linear Model achieving a slightly better result in terms of RMSE. However, the k-NN model showed higher RMSE, indicating difficulties in generalizing from the data. These models still had average of 12% margin of error. Advanced models like PCA, which reduces dimensionality by focusing on the most significant components, outperformed all others with the lowest RMSE and margin of error, proving to be the most accurate approach. This makes PCA the best choice for predicting resale prices. However, it is important to note that other economic factors, not captured by standard predictors, may have contributed to the extreme variations in the data. Two such factors will be discussed under the limitations section.

6. Limitations & Future Directions

A major limitation I faced was the large number of predictors, which made selecting the right ones and understanding their impact time-consuming. Data cleaning, particularly removing predictors with zero variance for the PCA model, added complexity. More efficient methods, such as Ridge Regression could be employed to improve feature selection and model performance. Furthermore, my initial error was to leave the resale price in the train and test set, which made my models perform exceptionally well. However, I realised it should be removed before plotting the model with the logged resale price as the response variable. Further improvements to improve model performance could include:

1. Including economic predictors: In 2019 the Enhanced CPF Housing Grant (EHG) was introduced offering SGD 80,000, to first time buyers (MND, 2019, September 10). This could potentially drive prices up in the following years, 2021, the year this data was collected, leading to outliers. The affordability index could serve as an indicator for this grant. Similarly, other indicators like inflation rates, property price index could also be added.
2. Extending time range: This data set was also collected during the post-pandemic period, which may have been volatile. Training and testing with subsequent stable years could increase robustness and accuracy of models, minimising risk of under of over predicting.

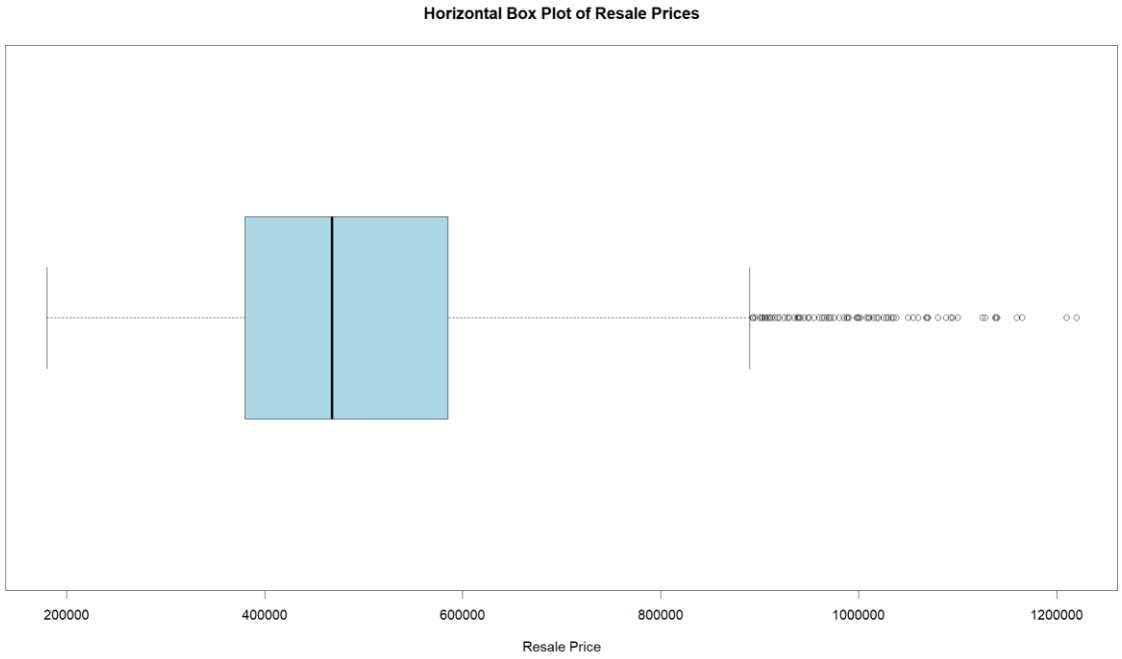
8. Citations

1. Ministry of National Development (MND): More affordable and accessible homes for Singaporeans: Enhanced CPF housing grant for First-Timers and higher income ceilings. (2019, September 10). https://www.mnd.gov.sg/newsroom/press-releases/view/more-affordable-and-accessible-homes-for-singaporeans-enhanced-cpf-housing-grant-for-first-timers-and-higher-income-ceilings-1
2. ChatGPT: I acknowledge the use of ChatGPT to modify and debug my code:
   - Used for removing data with 0 variances for the PCA analysis - Lines (383 – 413)

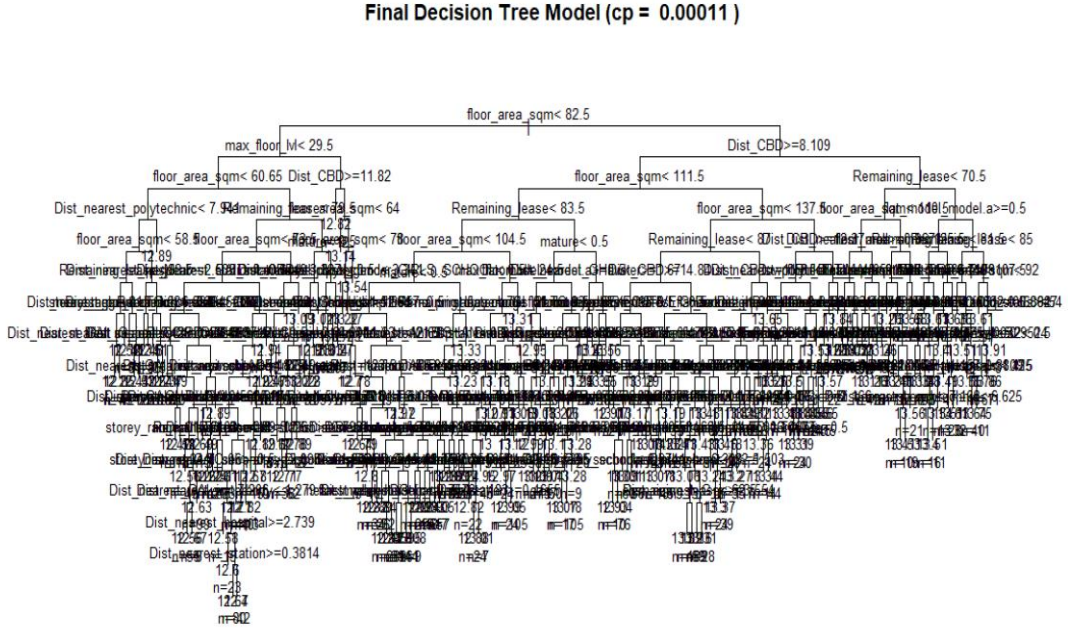
   The following prompts were given on 13/11/24:

   1. "Why is this error occurring? Error in La.svd(X) : infinite or missing values in 'x' In addition: Warning message: In FUN(X[[i]], ...) : Scaling with (near) zero standard deviation"
   2. "How do I remove data with 0 variances before commencing PCA analysis?"

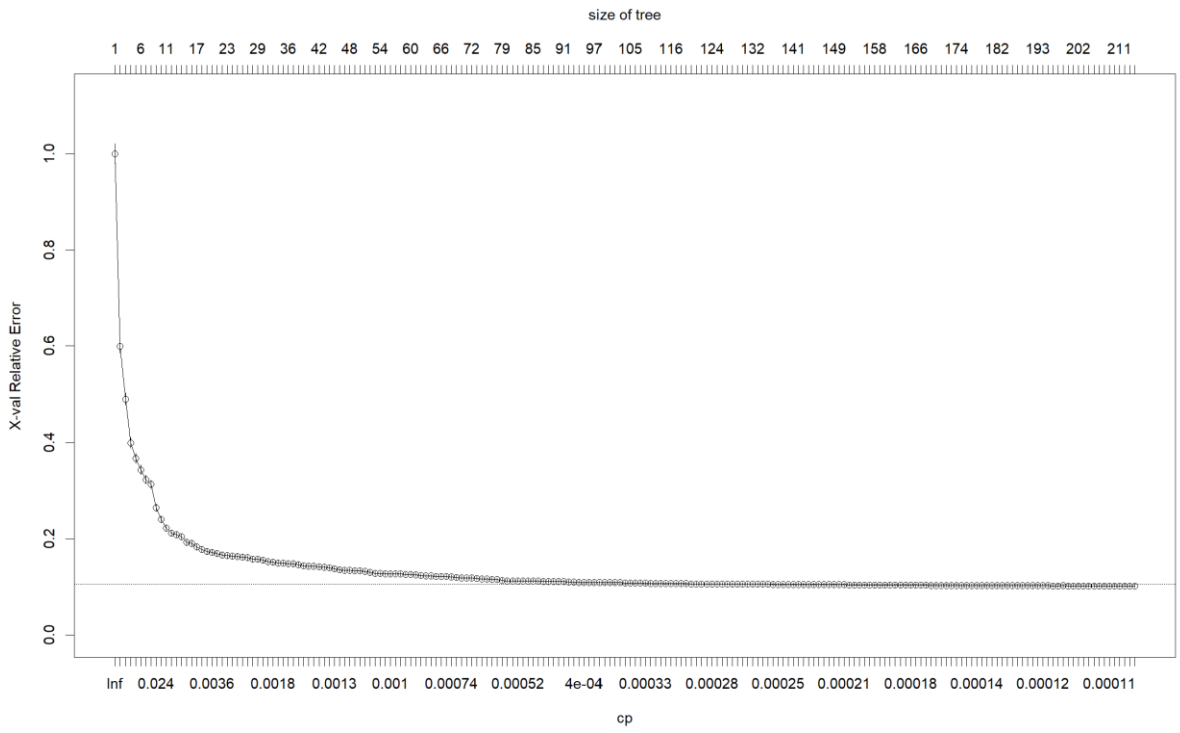
Appendix A



Appendix B



Appendix C



Appendix D

